University Institute of Engineering (UIE)

Subject:	**Computer Organisation and Architecture**
Student Name:	**Saanvi Aggarwal**
UID:	**25BAI10045**
Class/Section	**25BAI-601**

Scenario

A student runs a program that evaluates A = B + C, where B = 12 and C = 8. The CPU fetches, decodes, and executes the relevant machine instructions.

Problem Statement

Trace the operation of registers, ALU, control unit, and program counter during an addition instruction.

CASE STUDY QUESTIONS:

Q1. Describe the fetch, decode, and execute stages.

Ans.

Fetch : It is used to retrieve the instruction (A = B +C) from main memory using Program Counter. Then the instruction gets loaded into Instruction Register (IR) and PC increments to next instruction address .

Decode : The Control Unit decodes the instruction and identifies which operation is used to be performed and generates signal. It identifies the opcode (ADD operation) and the operands (R1 , R2 and R3) .

Execute : The ALU performs the operation on the required operands. The CU signals the ALU to read values from R1 and R2 , then performs addition . The output is written into destination register (R3).

Q2. What is the purpose of the Program Counter?

Ans.

The Program Counter stores the address of next instruction that needs to be executed. It is used to maintain the sequential program execution by incrementing automatically after each fetch step.

Q3. Which component performs arithmetic addition?

Ans.

The Arithmetic Logic Unit (ALU) performs arithmetic addition . It also performs other arithmetic and logical operations.

Q4. Why are CPU registers used?

Ans.

CPU registers are built directly into the processor core making high-speed. It is used to temporarily holds instructions, address, operands etc. providing fast storage inside CPU.

Q5. Where can the result be stored after execution?

Ans.

After execution the result gets stored either in destination CPU register (R3) or get transferred to main memory.

Assignment

Given R1 = 12 and R2 = 8, trace ADD R3, R1, R2. Draw a labeled basic CPU block diagram and annotate the path of both operands and the result.

Instruction that needs to be executed :

ADD R3 , R1 , R2 (This implies R3 <- R1 + R2)

Given:

R1 = 12

R2 = 8

Operation : R3 = R1 + R2

The CPU carries this entire operation by breaking it down into a sequence of instructions called the Instruction Cycle which involves:

i. Fetch
ii. Decode
iii. Execute

Step 1 : FETCH

This step is used to bring the instruction from main memory into the CPU.

The instruction is fetched from Program Counter (PC) , which contains the address of the next instruction.

Then, the address stored in PC is copied to Memory Address Register (MAR), which holds the memory address from which the instruction needs to be accessed.

The control unit sends a memory READ signal and main memory access the address of the instruction and retrieves ADD R3 , R1 , R2.

The instruction is then transferred to Memory Data Register (MDR) [holds the data or instruction being transferred to or from memory] and then to Instruction Register (IR) , which stores the current instruction.

The PC gets incremented and updated to next instruction.

Step 2 : DECODE

This step is used to understand what operation needs to be performed.

The instruction in IR gets examined by Control Unit.

The Control Unit breaks the instruction as follows:

ADD : Operation to perform

R3 : Destination Register

R1 AND R2 : Souce Register

The control unit then generates the signals to get the instruction executed

The Control Unit identifies R1 and R2 as the source registers and read its content

R1 = 12

R2 = 8

The value are then supplied to ALU (Arithmetic Logic Unit).

Step 3 : EXECUTE

This step is used to perform the actual arithmetic operation.

he Control Unit sends the ALU a control signal specifying the ADD operation.

The ALU performs using the binary adder

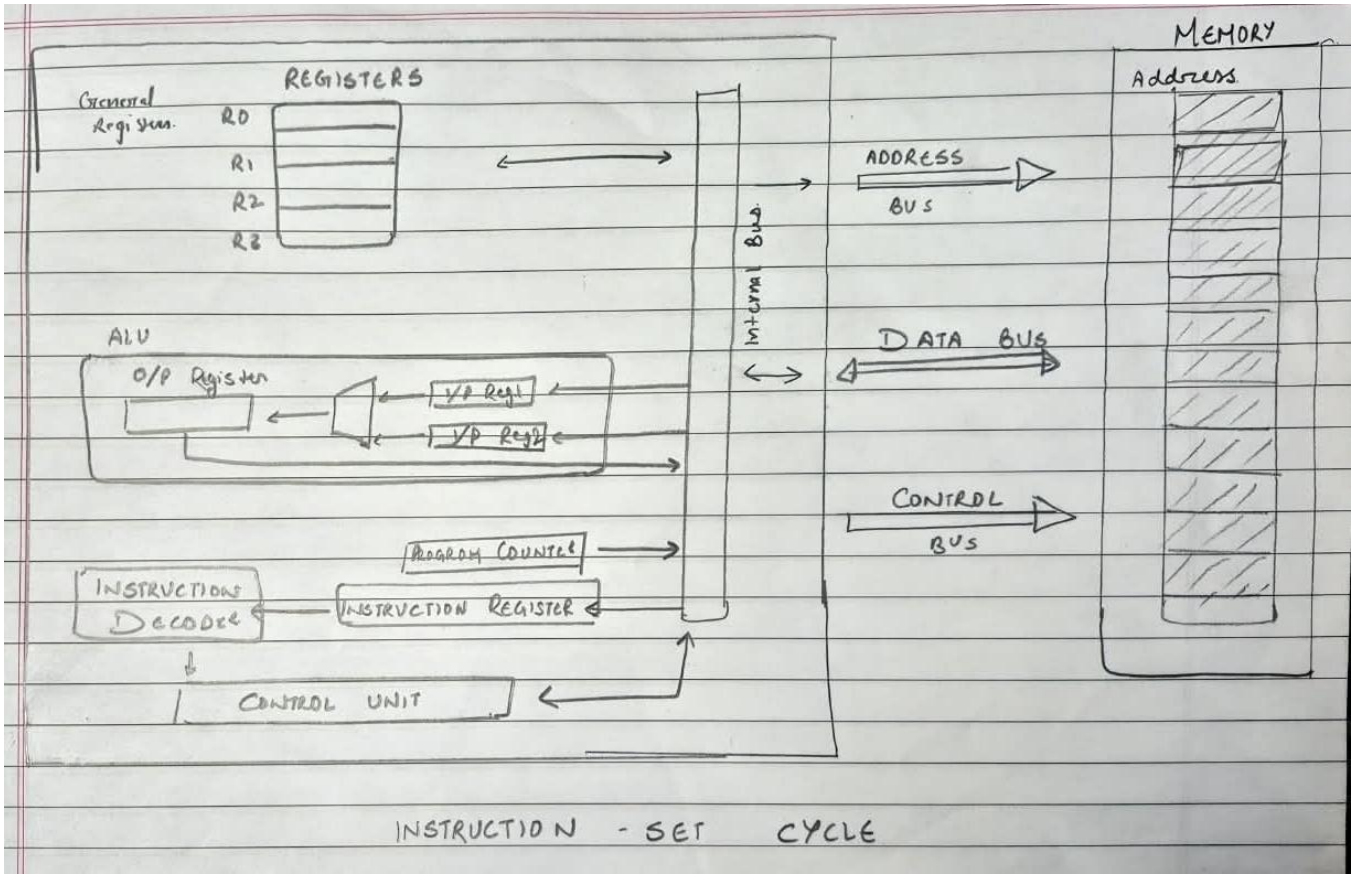
$0000\ 1100_2 + 0000\ 1000_2 = 0001\ 0100_2 \quad (20_{10})$

[12+8=20]

Step 4: Write-Back

The binary value of 20 from ALU output register is carried back to register file via data bus and gets updated in R3.

CHANDIGARH UNIVERSITY
Discover. Learn. Empower.
NAAC GRADE A+ ACCREDITED UNIVERSITY
QS UNIVERSITY RANKINGS ASIA 2025
AMONG ALL PRIVATE UNIVERSITIES IN INDIA



INSTRUCTION - SET CYCLE